Practical 2 - Understanding Foundation of Stats using Rating Dataset

1. Objective

The objective of this practical is to analyze the Teachers Rating dataset using fundamental statistical techniques. The practical focuses on identifying the type of data, calculating descriptive measures such as mean, median, minimum and maximum, generating descriptive statistics, visualizing the distribution of beauty scores using a histogram, comparing beauty scores across genders, and examining tenure status by gender.

2. Dataset Description

The Teachers Rating dataset contains information about professors and their course evaluations from top universities in the United States. The dataset consists of 463 observations and 19 variables. It includes demographic, course-related, evaluation, and employment-related characteristics of instructors.

Important variables used in this practical include **gender, beauty, eval, students, tenure, and tenured_prof.** The beauty variable represents the standardized beauty score of the instructor, while eval represents the teaching evaluation score. The students variable indicates the number of students in the course, and tenure indicates whether the instructor is tenured.

3. Tools and Technologies

The practical was implemented using Python for statistical analysis and data visualization. Pandas was used for loading, inspecting, and analyzing the dataset, while Matplotlib was used to create the histogram for the beauty variable. Visual Studio Code (VS Code) was used as the development environment for writing and executing the Python program. The final practical report was prepared using Google Docs.

Tool / Technology	Purpose

Python	Statistical analysis and data processing
Pandas	Dataset loading, manipulation, and statistical calculations
Matplotlib	Data visualization and histogram creation
VS Code	Writing and executing Python code
Google docs	Preparing and formatting the practical report

4. Question 1 -Time Series or Cross-Sectional Data

The Teachers Rating dataset is cross-sectional data.

Reason: The dataset contains observations of professors and their course evaluations at a particular point or period of observation. It does not contain a time or date variable that tracks the same observations over multiple time periods. Therefore, the dataset represents cross-sectional observations rather than time-series data.

Code:

```
# ============================================================

# QUESTION 1 : TYPE OF DATASET

# ============================================================



print("\n" + "=" * 70)

print("QUESTION 1 : TYPE OF DATASET")

print("=" * 70)



print("The Teachers Rating dataset is cross-sectional data.")
```

```
print("\nReason:")



print(
    "The dataset contains observations of professors/course evaluations "
    "and does not contain a time or date variable."
)



print(
    "Therefore, the dataset represents cross-sectional observations "
    "rather than observations tracked over time."
)
```

Output:

```
============================================================
QUESTION 1 : TYPE OF DATASET
============================================================
The Teachers Rating dataset is cross-sectional data.

Reason:
The dataset contains observations of professors/course evaluations and does not contain a time or date variable.
Therefore, the dataset represents cross-sectional observations rather than observations tracked over time.
```

5. Question 2 -Mean, Median, Minimum and Maximum

The students variable was analyzed to calculate its mean, median, minimum, and maximum values. The results are presented below.

Statistic	Value

Mean	36.6242
Median	23.0000
Minimum	5
Maximum	380

Code:

```
# ============================================================

# QUESTION 2 : MEAN, MEDIAN, MINIMUM AND MAXIMUM

# ============================================================



print("\n" + "=" * 70)

print("QUESTION 2 : STUDENTS VARIABLE STATISTICS")

print("=" * 70)



# Select the students column

students = df["students"]



# Calculate descriptive statistics

mean_students = students.mean()

median_students = students.median()

minimum_students = students.min()

maximum_students = students.max()
```

```
# Display the results

print("Mean number of students   :", mean_students)

print("Median number of students :", median_students)

print("Minimum number of students:", minimum_students)

print("Maximum number of students:", maximum_students)
```

Output:

```
================================================================================
QUESTION 2 : STUDENTS VARIABLE STATISTICS
================================================================================
Mean number of students   : 36.62419006479482
Median number of students : 23.0
Minimum number of students: 5
Maximum number of students: 380
```

Interpretation: The average number of students participating in the course evaluations is approximately **36.62**, while the median is **23** students. The minimum number of students is **5**, and the maximum is **380**. The mean is considerably higher than the median, indicating that some observations with very large numbers of students increase the average. This suggests that the distribution of the number of students is likely to be **right-skewed**.

6. Question 3 - Descriptive Statistics

Descriptive statistics were generated for the numerical variables in the Teachers Rating dataset. The table summarizes the count, mean, standard deviation, minimum, first quartile (25%), median (50%), third quartile (75%), and maximum for each numerical variable.

Variable	Count	Mean	Std. Dev.	Min	25%	Median	75%	Max
age	463	48.365	9.803	29	42	48	57	73
beauty	463	0.000	0.789	-1.450	-0.656	-0.068	0.546	1.970
eval	463	3.998	0.555	2.1	3.6	4.0	4.4	5.0
students	463	36.624	45.018	5	15	23	40	380
allstudents	463	55.177	75.073	8	19	29	60	581
prof	463	45.434	27.509	1	20	44	70.5	94
PrimaryLast	463	0.203	0.403	0	0	0	0	1
vismin	463	0.138	0.346	0	0	0	0	1
female	463	0.421	0.494	0	0	0	1	1
single_credit	463	0.058	0.235	0	0	0	0	1
upper_division	463	0.661	0.474	0	0	1	1	1
English_speaker	463	0.940	0.239	0	1	1	1	1
tenured_prof	463	0.780	0.415	0	1	1	1	1

Table 1. Descriptive statistics of the numerical variables in the Teachers Rating dataset.

Code:

```
# ============================================================

# QUESTION 3 : DESCRIPTIVE STATISTICS

# ============================================================



print("\n" + "=" * 70)

print("QUESTION 3 : DESCRIPTIVE STATISTICS")
```

```
print("=" * 70)



# Select numerical variables

numerical_columns = df.select_dtypes(include="number").columns



# Generate descriptive statistics

descriptive_statistics = df[numerical_columns].describe()



# Display the descriptive statistics table

print(descriptive_statistics)
```

Output:

```
============================================================
QUESTION 3 : DESCRIPTIVE STATISTICS
============================================================
              age        beauty        eval    students  allstudents  ...      female  single_credit  upper_division  English_speaker  tenured_prof
count  463.000000  4.630000e+02  463.000000  463.000000   463.000000  ...  463.000000     463.000000      463.000000       463.000000    463.000000
mean    48.365011  6.271140e-08    3.998272   36.624190    55.177106  ...    0.421166       0.058315        0.660907         0.939525      0.779698
std      9.802742  7.886477e-01    0.554866   45.018481    75.072800  ...    0.494280       0.234592        0.473913         0.238623      0.414899
min     29.000000 -1.450494e+00    2.100000    5.000000     8.000000  ...    0.000000       0.000000        0.000000         0.000000      0.000000
25%     42.000000 -6.562689e-01    3.600000   15.000000    19.000000  ...    0.000000       0.000000        0.000000         1.000000      1.000000
50%     48.000000 -6.801430e-02    4.000000   23.000000    29.000000  ...    0.000000       0.000000        1.000000         1.000000      1.000000
75%     57.000000  5.456024e-01    4.400000   40.000000    60.000000  ...    1.000000       0.000000        1.000000         1.000000      1.000000
max     73.000000  1.970023e+00    5.000000  380.000000   581.000000  ...    1.000000       1.000000        1.000000         1.000000      1.000000

[8 rows x 13 columns]
```

7. Question 4 - Histogram of Beauty Variable

A histogram was created to visualize the distribution of the beauty variable in the Teachers Rating dataset. The histogram shows the frequency of observations across different ranges of beauty scores.

Code:

```python
# ============================================================

# QUESTION 4 : HISTOGRAM OF BEAUTY VARIABLE

# ============================================================



print("\n" + "=" * 70)

print("QUESTION 4 : HISTOGRAM OF BEAUTY VARIABLE")

print("=" * 70)



# Create the histogram

plt.figure(figsize=(8, 5))



plt.hist(

    df["beauty"],

    bins=15,

    edgecolor="black"

)



plt.title("Distribution of Beauty Scores")

plt.xlabel("Beauty Score")

plt.ylabel("Frequency")



plt.tight_layout()



# Save the histogram
```

```
histogram_path = output_folder / "beauty_histogram.png"


plt.savefig(histogram_path, dpi=300)


# Display the histogram

plt.show()


print("\nHistogram saved to:")

print(histogram_path)


print("\nDistribution summary:")


print(
    "The beauty scores are centered approximately around zero "
    "and range from about -1.45 to 1.97."
)


print(
    "The distribution is fairly broad and somewhat irregular, "
    "with observations spread across both negative and positive "
    "beauty scores."
)
```

Output:

```
============================================================
QUESTION 4 : HISTOGRAM OF BEAUTY VARIABLE
============================================================

Histogram saved to:
/Users/anjanibharatchauhan/Documents/Developer/Statistics for DS/Statistics-for-Data-Science/Practical-2-Understanding-Foundation-of-Stats-using-Rating-Dataset
/output/beauty_histogram.png

Distribution summary:
The beauty scores are centered approximately around zero and range from about -1.45 to 1.97.
The distribution is fairly broad and somewhat irregular, with observations spread across both negative and positive beauty scores.
```

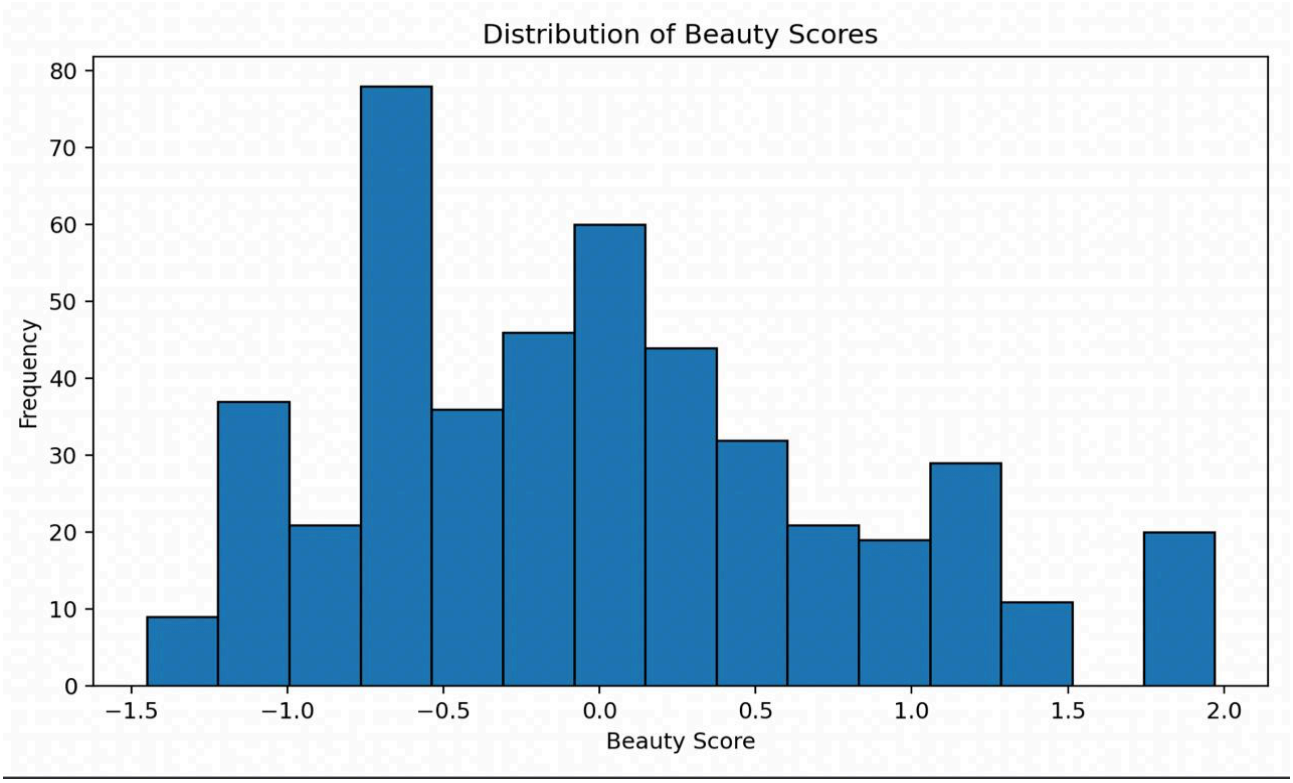


Figure 1. Histogram showing the distribution of beauty scores in the Teachers Rating dataset.

Interpretation: The histogram shows that the beauty scores are distributed over a range from approximately -1.45 to 1.97. The distribution is centered around zero, which is expected because the beauty variable is a standardized score. A large proportion of observations are concentrated between approximately -1.0 and 1.0, with fewer observations toward the extreme values. The distribution does not appear perfectly symmetric and shows some variation across the range of beauty scores.

8. Question 5 - Average Beauty Score by Gender

The average beauty score was compared between female and male instructors using the mean and standard deviation of the beauty variable. The results are summarized below.

Gender	Mean Beauty Score	Standard Deviation
Female	0.116	0.818
Male	-0.084	0.757

Table 2. Mean and standard deviation of beauty scores by gender.

Code:

```
# ============================================================

# QUESTION 5 : BEAUTY SCORE BY GENDER

# ============================================================



print("\n" + "=" * 70)

print("QUESTION 5 : BEAUTY SCORE BY GENDER")

print("=" * 70)



# Calculate mean and standard deviation of beauty

# separately for male and female instructors

beauty_by_gender = df.groupby("gender")["beauty"].agg(

    mean="mean",

    standard_deviation="std"

)
```

```
print("\nMean and Standard Deviation of Beauty Scores by Gender:")
print(beauty_by_gender)



# Extract individual values for interpretation
female_mean_beauty = beauty_by_gender.loc["female", "mean"]
female_std_beauty = beauty_by_gender.loc["female", "standard_deviation"]

male_mean_beauty = beauty_by_gender.loc["male", "mean"]
male_std_beauty = beauty_by_gender.loc["male", "standard_deviation"]

print("\nFemale instructors:")
print("Mean beauty score               :", female_mean_beauty)
print("Standard deviation              :", female_std_beauty)

print("\nMale instructors:")
print("Mean beauty score               :", male_mean_beauty)
print("Standard deviation              :", male_std_beauty)

print("\nComparison:")

if female_mean_beauty > male_mean_beauty:
    print("Female instructors have a higher average beauty score.")
```

```
elif male_mean_beauty > female_mean_beauty:

    print("Male instructors have a higher average beauty score.")

else:

    print("Male and female instructors have the same average beauty score.")
```

Output:

```
======================================================================
QUESTION 5 : BEAUTY SCORE BY GENDER
======================================================================

Mean and Standard Deviation of Beauty Scores by Gender:
             mean  standard_deviation
gender
female  0.116109             0.81781
male   -0.084482             0.75713

Female instructors:
Mean beauty score              : 0.11610907124450927
Standard deviation             : 0.8178096472554821

Male instructors:
Mean beauty score              : -0.08448223827351202
Standard deviation             : 0.7571299272149643

Comparison:
Female instructors have a higher average beauty score.
```

Interpretation: The mean beauty score for female instructors is **0.116**, while the mean beauty score for male instructors is **-0.084**. This indicates that female instructors have a higher average beauty score than male instructors in the dataset. The standard deviation is **0.818 for female instructors** and **0.757 for male instructors**, indicating some variation in beauty scores within both groups.

9. Question 6 - Tenure Status by Gender

Tenure status was analyzed by gender to compare the number and percentage of tenured instructors among female and male instructors. The results are summarized below.

Gender	Not Tenured	Tenured
Female	50	145
Male	52	216

Table 3. Tenure status counts by gender.

Percentage of Tenured Instructors by Gender

Female instructors: 74.36%
Male instructors: 80.60%

Code:

```
# ============================================================

# QUESTION 6 : TENURE STATUS BY GENDER

# ============================================================



print("\n" + "=" * 70)

print("QUESTION 6 : TENURE STATUS BY GENDER")

print("=" * 70)



# Create a count table of gender and tenure status

tenure_counts = pd.crosstab(

    df["gender"],
```

```
    df["tenure"]

)



print("\nTenure Counts by Gender:")

print(tenure_counts)




# Calculate the total number of instructors in each gender

gender_totals = df["gender"].value_counts()



# Calculate number of tenured instructors in each gender

tenured_counts = df[df["tenure"] == "yes"]["gender"].value_counts()



# Calculate percentage of tenured instructors within each gender

tenure_percentage = (

    tenured_counts / gender_totals * 100

)



print("\nPercentage of Tenured Instructors by Gender:")



print(

    tenure_percentage.rename("Percentage Tenured").to_frame()

)
```

```
# Extract percentages for interpretation
female_tenure_percentage = tenure_percentage["female"]
male_tenure_percentage = tenure_percentage["male"]

print("\nFemale instructors:")
print(
    "Percentage tenured:",
    female_tenure_percentage,
    "%"
)

print("\nMale instructors:")
print(
    "Percentage tenured:",
    male_tenure_percentage,
    "%"
)

print("\nComparison:")

if female_tenure_percentage > male_tenure_percentage:
    print(
```

```
        "The percentage of tenured instructors is higher among females."
    )
elif male_tenure_percentage > female_tenure_percentage:
    print(
        "The percentage of tenured instructors is higher among males."
    )
else:
    print(
        "The percentage of tenured instructors is the same for males and
females."
    )
```

Output:

```
================================================================================
QUESTION 6 : TENURE STATUS BY GENDER
================================================================================

Tenure Counts by Gender:
tenure  no  yes
gender
female  50  145
male    52  216

Percentage of Tenured Instructors by Gender:
        Percentage Tenured
gender
male             80.597015
female           74.358974

Female instructors:
Percentage tenured: 74.35897435897436 %

Male instructors:
Percentage tenured: 80.59701492537313 %

Comparison:
The percentage of tenured instructors is higher among males.
```

Interpretation: The results show that 145 female instructors and 216 male instructors were tenured. The percentage of tenured instructors was 74.36% among females and 80.60% among males. Therefore, the percentage of tenured instructors was higher among male instructors than female instructors in the dataset.

10. Conclusion

The Teachers Rating dataset was analyzed using fundamental statistical techniques.
The analysis identified the dataset as cross-sectional data and calculated important descriptive measures such as mean, median, minimum, maximum, and standard deviation.
The distribution of beauty scores was visualized using a histogram, showing that most observations were concentrated around the central range.
The comparison by gender showed that female instructors had a higher average beauty score, while male instructors had a higher percentage of tenured instructors.
Overall, the practical demonstrated how statistical methods can be used to summarize, analyze, and interpret real-world data.

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